Basic concepts of econometric estimation

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1 Basic concepts of econometric estimation

1.1 Population versus sample

The goal is to learn from a subset of observations (sample) about the whole population. Our assumption is that there is an unknown process that generates the data we have and that this process can be described by a probability distribution.

Any data we have as econometricians is the outcome of ***one*** *drawing in an experiment.*

To compute the characteristics of probability distribution, such as the expected values, variance, covariance, etc., we need the PDF, that is the whole sample space or population. To find out the average income of all people living in the city of Madrid at a given moment of time we need information on the whole population of Madrid. This is impossible.

We draw a *representative sample* from this population and compute the values we are interested in: e.g. expected value and variance.

Would the average income obtained from the sample be equal to the true average income in the population? Most likely not. But we should choose the best way to compute it.

Then, how to compute these values? We use special formulas (*recipes*) called **estimators**.

Example: We would like to find out the average income of all people living in the city of Madrid last week. We assume that this income is a random variable $X \sim N(\mu, \sigma^2)$. How to compute μ from a random sample of 100 incomes?

Let us propose two different estimators

- Sample mean: $\hat{\mu_1} = \bar{x} = \frac{x_1+x_2+\cdots+x_n}{n}$
- First-last value estimator: $\hat{\mu_2} = \frac{x_1+x_n}{2}$

We know intuitively that $\hat{\mu}_1$ is better. Why? What does it mean *better*?

In one draw $\hat{\mu}_2$ can be closer to the true μ than $\hat{\mu}_1$. But if we repeat our experiment, the difference comes out.

The reason is that the estimator $\hat{\mu}_1$ has better properties.

This is why we always look in econometric analyses for the estimators with best properties. In our case these properties are:

Random variable $X \sim N(\mu, \sigma^2)$

We cannot repeat the experiment in a real word, but we can do it by the use of simulation. Do not forget that in a real word there is only one sample (one draw).

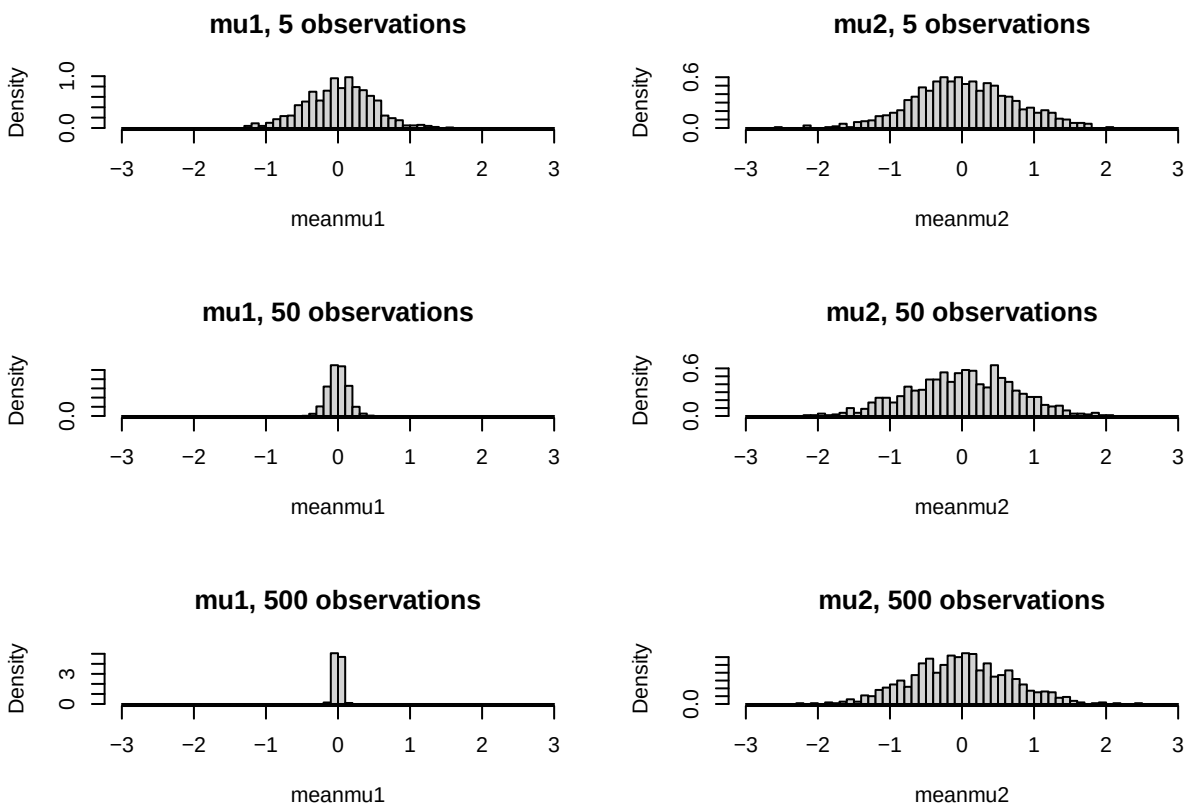
```
rm(list = ls(all = TRUE))                    # Clear workspace
Sys.setenv(LANG = "en")                      # Set language
setwd(".")                                   # Set working directory

set.seed(123456)
mat               <- matrix(1:6,3,2,byrow=T)
layout(mat)                                  # Set up the screen

intervals         <- c(seq(-3,3,0.1))
number.of.samples <- 1000
sample.sizes      <- c(5,50,500)

for(sample.size in sample.sizes){
  meanmu1 <- rep(0,number.of.samples)
  meanmu2 <- rep(0,number.of.samples)
```

```
  for (i in 1:number.of.samples){
   draws        <- rnorm(sample.size,mean=0,sd=1)
   meanmu1[i]   <- mean(draws)
   meanmu2[i]   <-((draws[1]+draws[sample.size])/2)
                                  }
 hist(meanmu1,
      breaks=intervals,
      main=paste("mu1,", sample.size,"observations"),
      prob=T)
 hist(meanmu2,
      breaks=intervals,
      main=paste("mu2,", sample.size,"observations"),
      prob=T)
}
```



Observed data is just one drawing in an experiment. The proposed estimator $\hat{\mu}$ for a particular dataset has a realized value called *estimation* (or estimate).

The variance of an estimator is often called its **sampling variance** since it is the variance associated with a sampling distribution. Remember the sampling variance is not a random variable; it is a constant, but it might be unknown.

We do not ask "how close is our estimation to the true μ?" in a particular sample. But we ask:

- How close is our estimation on average over various hypothetical drawings? I.e., what is the expected value of $\hat{\mu}$?(Note that $\hat{\mu}$ *is a random variable.*) Is our estimator unbiased, $E(\hat{\mu}) = \mu$?
- How variable is our estimation under repeated drawings? What is $Var(\hat{\mu})$. It indicates the precision of our estimator. -How does the variability change when our sample is arbitrarily large?

$$\lim_{N\to\infty} Pr(|\hat{\mu} - \mu| > \epsilon) = 0$$

Econometricians estimate parameters using OLS, ML, GLS, IV, etc. estimators. The goal is to use always an estimator with the best properties.

Some of these properties are:

- *Linearity* An estimator is said to be a *linear* estimator if it is a linear function of the sample observations. The sample mean is obviously a linear estimator because $\hat{\mu} = \bar{x} = \frac{x_1+x_2+\cdots+x_n}{n}$ is a linear function of the observations.
- *Unbiasedness*

$$E(\hat{\mu}) = \mu$$

- *Efficiency* If we consider only unbiased estimators of a parameter, the one with the smallest variance is called *efficient* estimator (relative efficiency). One way to compare estimators that are not necessarily unbiased is to compute the *mean squared error (MSE)* of the estimator. If $\hat{\mu}$ is an estimator of μ then the MSE of μ is defined as $MSE(\hat{\mu}) = E[(\hat{\mu}-\mu)^2]$. It can be shown that $MSE(\hat{\mu}) = V(\hat{\mu}) + [Bias(\hat{\mu})]^2$.
- *Best Linear Unbiased Estimator (BLUE)* If an estimator is linear, is unbiased, and has a minimum variance in the class of all linear unbiased estimators of a parameter, it is called a best linear unbiased estimator.
- *Consistency* The estimator $\hat{\mu}$ is said to be consistent if

$$\forall\epsilon \quad \lim_{N\to\infty} Pr(|\hat{\mu}-\mu| < \epsilon) = 1$$

That is $plim_{N\to\infty} = \mu$. Sufficient (not necessary) condition for consistency is: \if $\lim_{N\to\infty} E(\hat{\mu}) = \mu$ and $\lim_{N\to\infty} V(\hat{\mu}) = 0$ then $plim_{N\to\infty} = \mu$

1.2 General approaches to parameter estimation

Given a parameter θ appearing in a population distribution of a random variable Y, there are usually many ways to obtain unbiased and consistent estimator of θ.

1.2.1 Least Squares (LS)

The criterion of least square estimator is to make the sum of squared deviations as small as possible. The sample mean $\bar{X}$ is a LS estimator of the population mean μ of a random variable X. It can be shown that the value of m which makes the sum of squared deviations

$$\sum_{i=1}^{n}(X_i - m)^2$$

minimum is $m = \bar{Y}$. This estimator plays a major role throughout this course as it is the most applied estimator in the regression model framework.

Example in Excel: **Excel_EstimationLS_MuNormal.xlsx**

	A	B	C	D	E	F	G	H	I	J	K	L
1		Unknown for reseacher										
2		Observed by researcher										
3												
4		**Data generating process**										
5												
6		**Population parameters**					**Estimation**					
7		mean (mu)	0				mean (mu)	-0.00467928				
8		deviation (sigma)	1									
9												
10				**Hypothetical**								
11				**generated sample**			The criterion of least square estimator is to make the sum of squared deviations as small as possible.					
12		Uniform(0,1)	N(0,1)	**N(mu,sigma)**			**sum_i^N (x-meanhat)^2**	**194.175**				
13		0.590	0.228	**0.228**				0.054				
14		0.288	-0.560	**-0.560**				0.309				
15		0.373	-0.324	**-0.324**				0.102				
16		0.319	-0.470	**-0.470**				0.216				
17		0.673	0.448	**0.448**				0.205				
18		0.831	0.959	**0.959**				0.929				
19		0.666	0.429	**0.429**				0.188				
20		0.185	-0.895	**-0.895**				0.793				
21		0.251	-0.672	**-0.672**				0.445				
22		0.685	0.483	**0.483**				0.238				
23		0.658	0.406	**0.406**				0.168				
24		0.738	0.637	**0.637**				0.411				
25		0.771	0.742	**0.742**				0.557				
26		0.351	-0.382	**-0.382**				0.142				
27		0.902	1.296	**1.296**				1.691				
28		0.076	-1.434	**-1.434**				2.043				
29		0.923	1.427	**1.427**				2.051				

1.2.2 Maximum likelihood (ML)

This method requires the full specification of the distribution of the data (X_i) up to some parameters. The joint density of the observations, when considered as a function of these unknown parameters is called the likelihood function. We choose as our ML estimator those values for the unknown parameters that give us the highest likelihood, that is, those values which would generate our sample with highest probability.

Let us assume $X \sim N(\mu, \sigma^2)$. ML estimation of μ and σ^2 is obtained as follows. As $(X_1, X_2, \ldots, X_N)$ are independent, then

$$\begin{aligned} f(X_1, X_2, \ldots, X_N; \mu, \sigma^2) &= \Pi_{i=1}^N f(X_i; \mu, \sigma^2) \\ &= \Pi_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} exp\{-\frac{1}{2\sigma^2}(X_i - \mu)^2\} \ . \end{aligned}$$

Log-likelihood function is then defined as

$$\begin{aligned} \ln L(\mu, \sigma^2) &= \sum_{i=1}^N \ln f(X_i; \mu, \sigma^2) = \\ &= \sum_{i=1}^N \ln \left[(2\pi\sigma^2)^{-1/2} \exp\left\{ -\frac{1}{2\sigma^2} (X_i - \mu)^2 \right\} \right] = \\ &= -\frac{N}{2} \ln 2\pi - \frac{N}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^N (X_i - \mu)^2. \end{aligned}$$

The FOC's are given by:

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \begin{bmatrix} \frac{\partial \ln L(\theta)}{\partial \mu} \\ \frac{\partial \ln L(\theta)}{\partial \sigma^2} \end{bmatrix} = 0$$

that is

$$\frac{\partial \ln L(\theta)}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{N} (X_i - \mu) = 0$$
$$\frac{\partial \ln L(\theta)}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{N} (X_i - \mu)^2 = 0$$

The MV estimator of μ and σ^2 is the solution of the above equation system which can be rewritten as:

$$\sum_{i=1}^{N} (X_i - \hat{\mu}) = 0$$
$$N - \frac{1}{\hat{\sigma}^2} \sum_{i=1}^{N} (X_i - \hat{\mu})^2 = 0$$

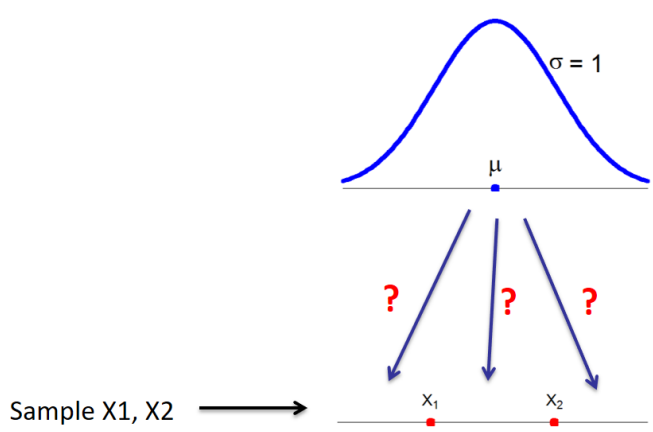
which leads to

$$\hat{\mu} = \frac{\sum_{i=1}^{n} X_i}{n}$$
$$\hat{\sigma}^2 = \sum_{i=1}^{n} \frac{(X_i - \bar{X})^2}{n}$$

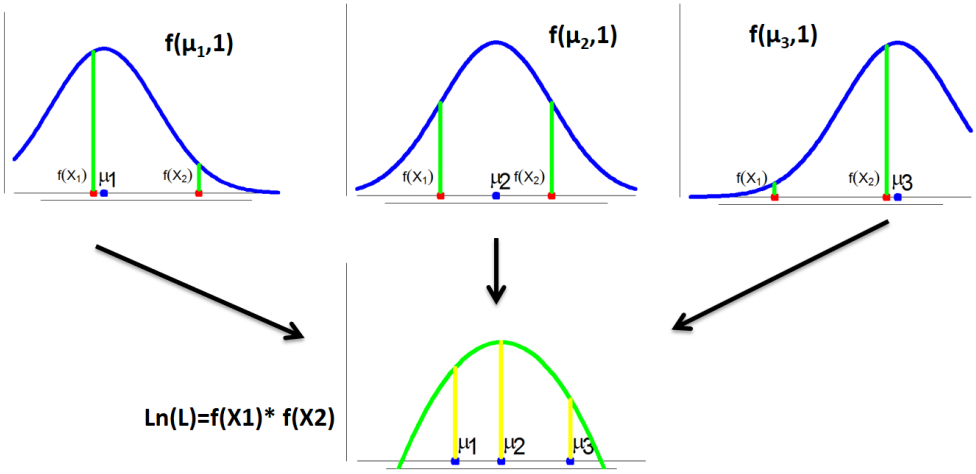
1.2.2.1 Estimation of μ in $N(\mu, 1)$ case

Criterion:

We choose as our ML estimator those values for the unknown parameters that give us the highest likelihood, that is, **those values which would generate our sample with highest probability**.

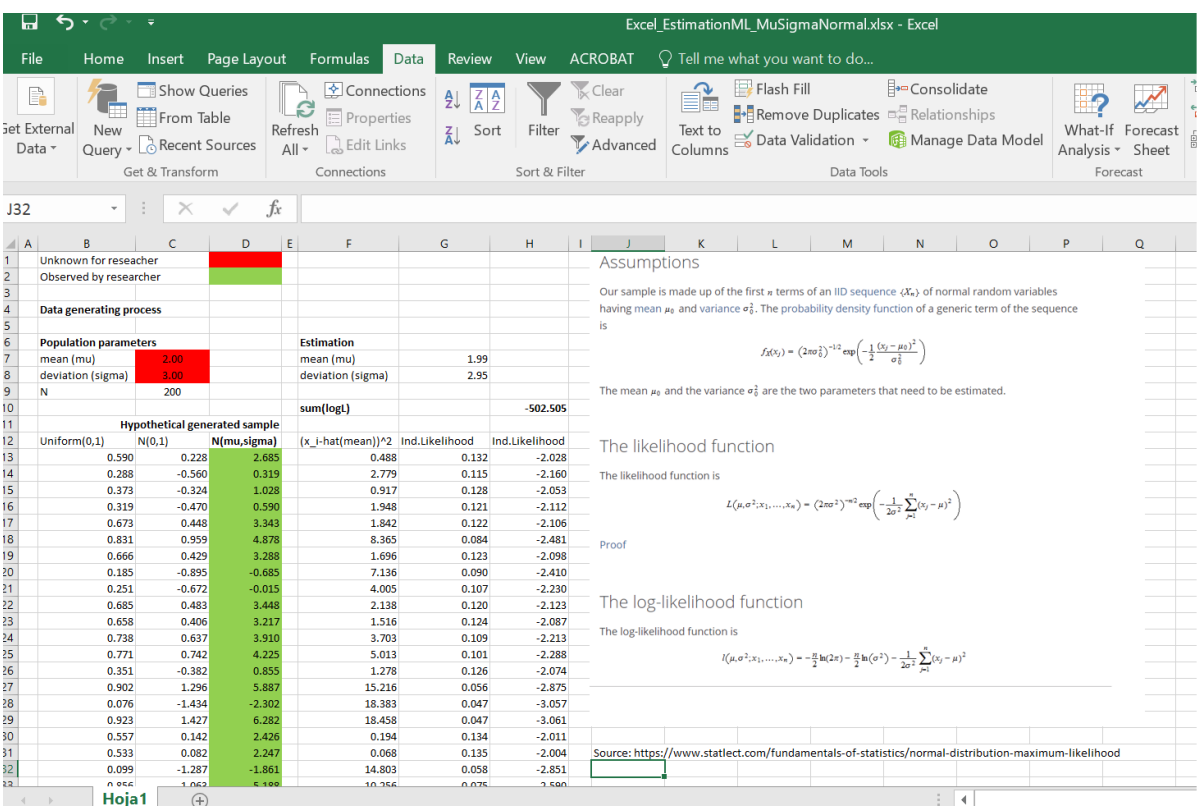


We choose as our ML estimator those values for the unknown parameters that give us the highest likelihood, that is, **those values which would generate our sample with highest probability**.



1.2.2.2 Estimation of μ and σ in $N(\mu, \sigma^2)$ case by ML.

Example in Excel: **Excel_EstimationML_MuSigmaNormal.xlsx**



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